

NAVYA THELLAPATI

Tampa, FL | +1 (813) 363-2834 | navyachowdary.thellapati@gmail.com | [LinkedIn](#)

PROFESSIONAL SUMMARY

Software Engineer with 3+ years of experience building backend applications, REST API integrations, workflow automation, and data validation solutions for enterprise financial services environments. Strong experience supporting account servicing, billing, payment, and internal operations workflows through reliable code, SQL-driven validation, API testing, and CI/CD delivery. Works closely with business analysts, QA teams, database teams, and product owners to turn requirements into tested, maintainable software updates. Brings hands-on project experience in backend and API-driven application workflows, AI-assisted document search, healthcare systems, and responsible machine learning, with a practical focus on building stable software that supports real business needs. Currently pursuing an M.S. in Computer Science at the University of South Florida.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, SQL, Bash

Backend & API Development: FastAPI, Flask, RESTful APIs, Microservices, SQLAlchemy, OpenAPI, JSON, XML, JWT, API Integration

Frontend Development: React.js, HTML5, CSS3, Responsive Web Design, Component-Based Development

Databases: PostgreSQL, SQL Server, Relational Database Design, SQL Query Optimization

Cloud, DevOps & CI/CD: Docker, Jenkins, Git, GitHub, GitHub Actions, Azure DevOps, Linux, CI/CD Pipelines

Testing & Debugging: Postman, API Testing, Unit Testing, Regression Testing, Test Data Validation, Debugging, Defect Tracking

AI-Integrated Software Development: RAG, LangChain, OpenAI API, Llama 3, ChromaDB, Vector Databases, Semantic Search, Sentence-Transformers, Prompt Engineering, LLM API Integration, Source Attribution

ML & Responsible AI: TensorFlow, scikit-learn, Pandas, NumPy, Neural Networks, Model Evaluation, Bias Detection, Fairness Validation

Software Engineering: Agile, Scrum, SDLC, Object-Oriented Programming, Data Structures, Algorithms, Code Reviews, Jira, Confluence

PROFESSIONAL EXPERIENCE

Associate Software Engineer

Jun 2021 - Jul 2024

Accenture Solutions | Client: Edward Jones

- Automated client account servicing, billing review, payment validation, and internal operations workflows using Python and SQL, reducing 300+ recurring manual checks and improving processing efficiency by 28%.
- Integrated 20+ internal account, billing, customer profile, and payment services through Python, REST APIs, microservice-style API logic, JSON, XML, and API integration workflows, supporting 6K+ daily financial records with fewer manual handoffs.
- Maintained Python/SQL ETL routines that moved client, account, transaction, and billing data into downstream reporting tables, reducing recurring load errors by 22% for operations teams.
- Optimized SQL reconciliation queries across PostgreSQL and SQL Server to validate customer accounts, payment records, billing entries, and test datasets, reducing pre-release data mismatches by 25%.
- Contributed to reusable backend automation modules using Python, SQLAlchemy, exception handling, and structured logging, making recurring extraction, validation, and loading workflows easier to maintain.
- Streamlined release consistency by packaging Python applications through Jenkins CI/CD pipelines, Git, Azure DevOps work items, Linux build environments, and deployment checks, supporting 12+ release validations.
- Strengthened backend reliability by adding input validation, retry handling, debugging support, and transaction-level logs, helping engineering teams trace failed API calls, incomplete records, and payment exceptions faster.
- Collaborated with business analysts, QA engineers, database teams, and product owners through Agile/Scrum, Jira, Confluence, defect tracking, and code reviews to deliver tested software updates.

PROJECTS

DocuMind - AI Document Q&A Assistant

- Implemented a RAG-based semantic search application using Python, FastAPI, LangChain, ChromaDB, Llama 3, and OpenAI API, helping users query 1,000+ documents with source-attributed answers.
- Configured semantic chunking, Sentence-Transformers embeddings, ChromaDB vector storage, retrieval logic, and prompt engineering templates to improve answer relevance and reduce unsupported responses by 35%.
- Exposed OpenAPI-documented REST endpoints for document upload, indexing, search, Q&A, feedback, and chat history, then packaged the backend with Docker and GitHub Actions for repeatable testing and deployment readiness.

MyChart - Full-Stack Healthcare Patient Portal

- Built a responsive healthcare portal using React.js, TypeScript, HTML5, CSS3, Flask/FastAPI, PostgreSQL, and SQLAlchemy to support appointments, lab results, medications, visit history, billing, and provider messaging.
- Developed a FastAPI-based AI chatbot by integrating OpenAI API, REST endpoints, JSON request handling, and guided response flows to help users navigate appointments, billing, medications, test results, FAQs, and portal actions.
- Implemented JWT authentication, role-based access, OpenAPI validation, Postman API testing, unit tests, regression checks, and PostgreSQL verification queries to secure workflows and improve prototype task completion by 30%.

Fairness in Income Prediction & Responsible AI

- Trained a three-layer neural network on 48,842 UCI Adult Census records using TensorFlow, scikit-learn, Pandas, and NumPy, reaching 85% accuracy and 0.90 ROC AUC.
- Assessed race and gender prediction fairness using bias detection, fairness validation, 80% Rule analysis, and model evaluation metrics, identifying a 12% disparity for mitigation review.
- Tuned the classification workflow with TensorFlow, scikit-learn, Pandas, NumPy, debiasing techniques, and validation checks, maintaining 82% accuracy while making model outcomes more explainable for responsible AI use.

EDUCATION

Master of Science in Computer Science

Aug 2024 - May 2026

University of South Florida | Tampa, FL

CERTIFICATIONS

- Meta Back-End Developer Professional Certificate - **Coursera**
- IBM Full Stack Software Developer Professional Certificate - **Coursera**
- Google IT Automation with Python Professional Certificate - **Coursera**
- Meta Front-End Developer Professional Certificate - **Coursera**
- GitHub Actions for CI/CD: Build, Test, and Deploy - **LinkedIn Learning**